Research on Autonomous-Driving Object Detection with YOLOv8 Improved by a RepViT Backbone

Jintao Lai
College of Information Engineering
Henan University of Science and Technology
Luoyang, China
231404010112@stu.haust.edu.cn

2026 5th International Conference on Electronics Technology and Artificial Intelligence (ETAI) | 979-8-3315-7820-6/26/$31.00 ©2026 IEEE | DOI: 10.1109/ETAI68332.2026.11485359

***Abstract*—Balancing detection accuracy and real-time requirements under constrained onboard compute remains a key challenge for autonomous-driving perception. This paper explores a RepViT-backbone variant of Ultralytics YOLOv8 by replacing the original backbone with RepViT, which supports structural re-parameterization (multi-branch training with single-branch inference). To interface RepViT with the multi-scale feature pathway of YOLOv8, we design a RepViTStageYOLO module and implement the necessary engineering adaptations within the Ultralytics framework. Experiments on the KITTI dataset show that the proposed model achieves competitive detection accuracy, reaching an mAP@0.5 of 0.892, with strong performance on vehicle categories (Car AP: 0.969). This paper focuses on accuracy evaluation; efficiency metrics such as Params, FLOPs/GFLOPs, and FPS are not reported and will be investigated in future work for a complete assessment of deployment efficiency.**

Keywords—Autonomous driving; Object detection; YOLOv8; RepViT; Structural re-parameterization; Efficient backbone;

I. Introduction

With autonomous driving and intelligent transportation systems emerging as the focus of global technological competition, object detection technology based on computer vision serves as the key "eye" for perception, and its performance is directly related to driving safety and reliability. However, existing high-performance detection models are often accompanied by enormous computational overhead, making it hard to support real-time inference on on-board edge devices with limited computing power and power consumption.

To mitigate these overheads, various lightweight strategies have been explored. For instance, ECA-Net [10] introduces an efficient channel attention mechanism to enhance feature representation with minimal cost, while hierarchical structures like Swin Transformer [11] and structural re-parameterization techniques such as RepVGG [12] significantly optimize the accuracy-efficiency trade-off. What's more, the latest YOLOv8 algorithm still suffers from computational redundancy in its backbone network under extremely constrained environments.

Recent studies have actively sought to improve the YOLO framework for diverse applications; examples include Qi's [13] work on attention-based YOLOv5, Zhang's [14] infrared detection variant, and Zhong's [15] real-time detection model using Rep structures. Furthermore, specialized frameworks like EdgeTrim-YOLO [16] and HCLT-YOLO [17] highlight the importance of model pruning and hybrid CNN-Transformer architectures for complex traffic scenarios.To address this challenge, this study explores introducing the RepViT network, which combines ViT-style global modeling with a structural re-parameterization mechanism, into the YOLOv8 framework. We achieve precise alignment of the four feature maps from C2 to C5 by designing the adaptive RepViTStageYOLO module. After resolving engineering issues such as module registration and multi-process training conflicts, we evaluate the proposed model on the KITTI dataset and show that it maintains competitive detection accuracy (mAP@0.5 = 0.892, Car AP = 0.969). While the RepViT design is motivated by inference efficiency via re-parameterization, this paper primarily reports accuracy results; a dedicated efficiency benchmark (Params/FLOPs/FPS) will be conducted in future work.

II. Relevant Theories and Technical Foundations

A. The principle of the YOLOv8 algorithm

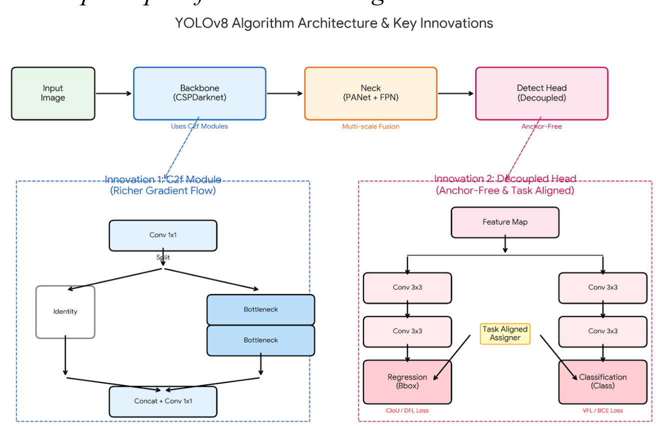

Fig. 1. YOLOv8 Concept Diagram

YOLOv8 has achieved a remarkable boost in detection performance with a full revamp of its backbone network, detection head and sample assignment mechanism [1]. Its backbone network centers on the C2f module. This module evolves from the ELAN structure of YOLOv7. It adds flexible branched feature flows and cross-stage connections. This effectively strengthens the ability to model the interaction between shallow detailed information and deep semantic information. It also cuts down computational redundancy and eases the gradient vanishing problem in deep networks.The overall architectural visualization of these components is presented in Fig. 1.

YOLOv8 adopts a Decoupled Head design at the prediction stage[2]. It explicitly separates classification and regression tasks. It optimizes the prediction of category probabilities and the regression of bounding box geometric parameters through independent branches respectively. This solves the gradient competition problem in traditional coupled heads effectively. It also improves localization accuracy and training stability in complex scenarios.

Besides, the algorithm fully introduces the Anchor-Free detection mechanism [3][4]. It directly regresses the offset between the feature map pixels and the target boxes. This reduces the reliance on prior anchor boxes. It also combines with the Task-Aligned Assigner sample assignment strategy. The strategy comprehensively considers classification scores and regression quality to dynamically select positive samples. This design shows stronger robustness and generalization ability. It performs better when dealing with small objects like pedestrians and cyclists in the KITTI dataset, as well as scenes with dense object distribution.

B. RepViT Lightweight Network

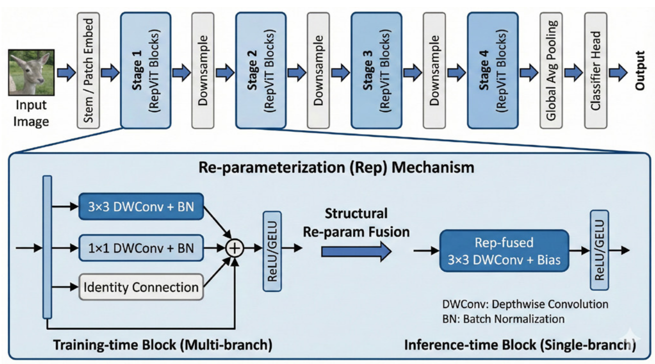

Fig. 2. RepViT Lightweight Network Architecture Diagram

The RepViT network is designed to balance representation capacity and computational practicality by introducing structural re-parameterization and a ViT-inspired modeling paradigm. It adopts a multi-branch parallel structure during training to enrich feature learning with diverse receptive fields and improve optimization stability. During inference, the trained multi-branch structure can be mathematically fused into an equivalent single-path convolutional form, which is expected to be more hardware-friendly. This study does not provide a quantitative efficiency analysis (e.g., FLOPs/latency) and leaves such benchmarking for future work.The detailed architectural hierarchy and components of RepViT are illustrated in Fig. 2.

In addition, RepViT deeply integrates the core modeling ideas of ViT. It introduces the concept of Token Mixing and depth-wise separable convolution. This simulates the global information interaction capability of Transformer. It effectively makes up for the deficiency of traditional lightweight convolutional networks in modeling long-range dependencies [5]. Its overall architecture adopts a hierarchical feature extraction strategy similar to Swin Transformer [11]. It realizes the collaborative expression of local details and high-level semantics through layer-by-layer downsampling and channel dimension enhancement. This strengthens the model's multi-scale feature modeling capacity and also makes it. It can naturally adapt to the feature fusion requirements of detection frameworks such as YOLOv8. Combined with its parameter-efficient design and compatibility with modules such as SE and ECA [10], RepViT provides a compact backbone option while preserving expressive feature representations.

C. Evaluation Metrics

To conduct a comprehensive and objective evaluation of the proposed RepViT-based YOLOv8 improved model for object detection performance on the KITTI dataset, this paper selects a variety of commonly used and representative evaluation metrics for quantitative analysis of the model's performance. Specifically, the experiment uses metrics such as Precision, Recall, F1-score and mean Average Precision (mAP)[6][7]. It makes a comprehensive evaluation of the model from multiple perspectives, including classification accuracy, object recall capability and overall detection performance.

At the same time, combined with the changes of Loss curves such as bounding box regression loss, classification loss and distributed focal loss during the training process, the paper further analyzes the model's convergence speed, training stability and optimization effect. This multi-metric, multi-dimensional evaluation method can more comprehensively reflect the model's detection performance and practical application potential in real road scenarios. In this work, we focus on accuracy-related metrics; efficiency metrics such as Params, FLOPs/GFLOPs, and FPS are not reported and will be evaluated in future work.

III. DESIGN OF IMPROVED YOLOV8 ALGORITHM BASED ON REPVIT

This paper proposes an improved YOLOv8 detection algorithm based on a RepViT backbone. The model preserves the original end-to-end one-stage detection paradigm of YOLOv8 while replacing the backbone feature extractor with RepViT and introducing the RepViTStageYOLO module to produce multi-scale feature maps (C2–C5) that match the subsequent FPN+PAN pathway.

This design follows the research method of optimizing YOLO for specific tasks. Qi [13] improved YOLOv5 with attention mechanisms, and Zhang [14] created a YOLOv8 version for infrared detection. Other researchers like Zhong [15] have also tested real-time Rep structures. Similar to the lightweight goals in EdgeTrim-YOLO [16] and HCLT-YOLO [17], our work focuses on the integration of RepViTStageYOLO.

The detection head retains the Decoupled Head and Anchor-Free mechanisms to maintain stable optimization and localization quality. Our experiments mainly evaluate detection accuracy on KITTI; a complete efficiency benchmark (Params/FLOPs/FPS) for deployment-oriented analysis is left for future work [8].

A. Integration of the RepViT Backbone Network

Given the stringent requirements for computing resources in on-vehicle embedded deployment, this paper introduces the RepViT lightweight network to restructure the backbone network of YOLOv8 in a targeted manner. It aims to reduce the parameter scale and computational complexity. To adapt to the modular parsing mechanism of YOLOv8 and the demand for multi-scale feature output, this study divides RepViT into multiple stages and packages them into the RepViT StageYOLO module. It stacks reparameterizable RepViT Blocks, ensuring that the output feature maps can accurately connect to the subsequent detection head interfaces.

This backbone network continues the layer-by-layer downsampling strategy. It also enhances the global modeling capability through the Token mixing mechanism. And it uses structural re-parameterization to equivalently fuse the multi-branch structure into a single-path convolutional form at inference time, which is expected to improve hardware-friendliness. A systematic deployment-oriented evaluation of efficiency (Params/FLOPs/FPS) is beyond the scope of this paper and will be conducted in future work.

B. *Implementation of Core Modules*

The implementation of the core modules of the RepViT-YOLO model focuses on the adaptive transformation and engineering integration of RepViTBlockYOLO. In the training phase, this module strengthens the modeling of local spatial features through a reparameterizable multi-branch structure, and improves the richness and robustness of feature expression by utilizing different receptive fields. In the inference phase, it equivalently fuses the multi-branch structure into a single-path convolution via structural reparameterization technology, which is intended to simplify the inference-time computation via structural re-parameterization; however, this paper does not report dedicated latency/FPS measurements, and such evaluation will be included in future work, realizing the design concept of complex training and simple inference. The RepViTStageYOLO module constructed on this basis replaces the original C2f module of the YOLOv8 backbone network by stacking basic units. Through a hierarchical feature extraction mechanism, it gradually enhances semantic information and preserves fine-grained details as the network deepens, providing more discriminative multi-scale inputs for the subsequent detection head.

To ensure the stability of the algorithm under the Ultralytics framework, this paper completes the dynamic registration of custom modules at the framework entry by extending the module management process, enabling RepViTStageYOLO to be correctly identified by the model parser and supporting instantiation via YAML configuration files, thus guaranteeing the compatibility of training and inference logic. In addition, an interface for attention mechanisms is reserved in the module design, allowing for the convenient insertion of lightweight modules such as SE or ECA in the follow-up, which lays a solid engineering foundation for further enhancing channel modeling capability and conducting performance optimization experiments.

IV. Experimental Environment and Dataset Configuration

A. *Experimental Environment*

To verify the effectiveness of the proposed algorithm, this experiment was conducted on a high-performance computing platform. The software environment was built on the Windows operating system, with the model constructed using the PyTorch deep learning framework and secondary development carried out based on the Ultralytics open-source library.

TABLE I. Experimental Configurations

Configuration Item	*Relevant Configurations*
GPU Hardware	NVIDIAGeForceRTX4060LaptopGPU
Deep Learning Framework	PyTorch2.8.0
Input Size	640×640
Optimizer	SGD (momentum=0.9，weightdecay=0.0005)
Training Epochs	100epochs
Batch Size	16

B. *Dataset Processing*

This study adopts the KITTI object detection dataset as the primary evaluation benchmark. The KITTI dataset provides high-quality, multi-category, and multi-scale real road data for object detection tasks in autonomous driving scenarios, serving as an authoritative benchmark for evaluating the performance of detection algorithms. Collected by on-vehicle cameras, the dataset contains complex traffic conditions and illumination variations [9]. To adapt to the YOLOv8 detection framework, this study conducts standardized processing on the original category system and label format of the KITTI dataset. Aiming at the significant differences between KITTI annotations and the COCO system, this study completely retains eight representative traffic object categories including Car, Van, Truck, Pedestrian, Person_sitting, Cyclist, Tram and Misc. These categories are renumbered as continuous integer labels in accordance with YOLO specifications, realizing category reorganization while avoiding information loss. In terms of format conversion, the original text annotations containing redundant 3D information and pixel coordinates are uniformly mapped to the YOLO five-tuple format composed of category index, normalized center coordinates, and width-height. This ensures full compatibility of the data with the Ultralytics training process while preserving the original annotation accuracy, laying a standardized data foundation for the efficient full-scale training of subsequent models. After completing data preprocessing, this study compiles a YAML dataset configuration file compliant with YOLOv8 specifications to realize automatic loading and parsing of the KITTI dataset. The configuration file specifies the dataset root directory, the path division of training and validation sets, and defines the semantic names of the eight detection object categories in sequence, ensuring the consistency between model output indices and actual categories. Through this standardized configuration method, the training program can automatically execute sample loading and label matching without additional script intervention. While improving the standardization and reproducibility of the experimental process, it also provides a good general interface for the subsequent expansion of other autonomous driving datasets.

V. Experimental Results and Analysis

A. *Training Convergence Analysis*

To verify the training stability and convergence characteristics of the RepViT-based improved YOLOv8 model on the KITTI dataset, this paper analyzes the variations in the loss functions and key performance metrics during the training process. Fig.3 presents the curves of box loss, cls loss and dfl loss of the model on the training and validation sets with the number of training epochs, and also shows the evolution trends of Precision, Recall and mAP metrics.

As can be seen from the changes in loss functions in Fig.3, all loss values of the model drop rapidly in the early stage of training, indicating that the network can effectively learn the spatial location and category information of targets. With the increase of training epochs, the decline rate of the loss functions gradually slows down and stabilizes in the middle and late stages without obvious oscillation or rebound, which demonstrates a relatively stable training process. The loss curves of the validation set and the training set show an overall consistent trend with no obvious divergence, indicating that the model does not suffer from severe overfitting and has good generalization ability.

In terms of performance metrics, Precision and Recall increase gradually with the number of training epochs and stabilize in the later stage, which proves that the model has been effectively optimized in both false detection control and target coverage capability. Meanwhile, the curves of mAP@0.5 and mAP@0.5:0.95 show a continuous rising trend, further verifying that the detection performance of the model under different IoU threshold conditions is constantly improved. Based on the comprehensive analysis of the changes in loss functions and performance metrics, it can be concluded that the proposed improved model has favorable training convergence and stability on the KITTI dataset.

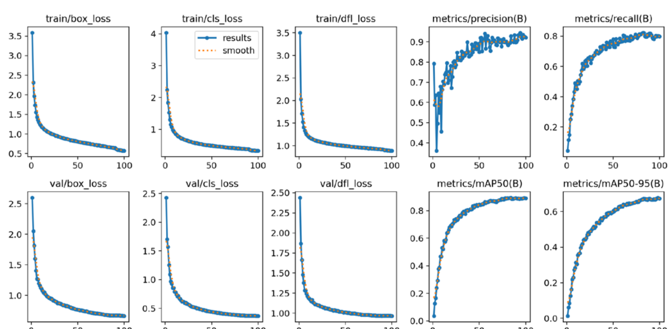

Fig. 3. Variation Curves of Loss Functions and Performance Metrics of the Improved YOLOv8 Model in Training and Validation Phases

B. Overall Detection Performance Evaluation

To comprehensively evaluate the object detection performance of the improved YOLOv8 model on the KITTI dataset, this paper conducts an analysis from multiple perspectives including the Precision-Recall curves, F1-Confidence curves and the overall mAP metric. Fig.4 shows the Precision-Recall curves of the model for each object category, and present the variations of F1, Precision and Recall with the confidence threshold respectively.

It can be observed from the PR curves in Fig.4 that the curves for vehicle targets such as Car, Van and Truck are generally located in the higher region with a large area under the curve, indicating that the model has a strong detection capability for targets with distinct structural features and relatively stable scales. The overall average PR curve maintains a high level, reflecting the superior comprehensive performance of the improved model in multi-category detection tasks.

The F1-Confidence curve indicates that the model achieves an optimal F1 value at a confidence level of approximately 0.5, which means the model realizes a relatively ideal balance between Precision and Recall within this threshold range. This result provides a basis for the selection of confidence threshold in the practical application of the model.

The Precision of the model continuously improves with the increase of the confidence threshold; combined with the Recall-Confidence curve in Fig.4, it can be seen that Recall gradually decreases with the increase of the threshold, reflecting the typical trade-off between detection precision and recall rate. Under the condition of high confidence, the prediction results of the model are more reliable, but some targets may be filtered out; under the condition of low confidence, the model has a higher target coverage capability, but the risk of false detection increases accordingly. On the whole, the improved model exhibits a relatively smooth performance variation under different threshold settings, indicating that its prediction results have good stability.

C. Category-level Performance Comparative Analysis

To further analyze the model's detection capability for different object categories, a category-level performance evaluation was conducted on all object categories in the KITTI dataset. From the PR curves and mAP values of each category, it can be seen that the model performs particularly well on vehicle targets such as Car, Van and Truck, with the PR curves of these categories being relatively smooth overall and their mAP values remaining at a high level. This is mainly attributed to the advantages of the RepViT backbone network in multi-scale feature modeling, which enables the model to better capture the structural features of vehicle targets.

For pedestrian and non-motor vehicle categories including Pedestrian and Cyclist, the model's detection performance is relatively lower than that for vehicle categories, but remains within an acceptable range overall. Since such targets in the KITTI dataset are generally characterized by small scales, large pose variations and severe occlusions, they pose higher requirements for detection models. The experimental results show that despite certain challenges, the model with the introduction of RepViT still exhibits good robustness in small target scenarios, and the Recall curves do not decline prematurely, indicating that the model can stably detect most real targets.

In addition, categories such as Person_sitting and Misc show relatively low detection performance due to their relatively small sample sizes and insufficiently concentrated category features. This phenomenon is closely related to the inherent category distribution characteristics of the dataset, and also reflects from another perspective that there is room for improvement by further introducing attention mechanisms or category reweighting strategies in category-imbalanced scenarios.

Based on the comprehensive results of the category-level analysis, it can be concluded that the RepViT-based improved YOLOv8 model has strong detection capability for major object categories and good applicability in complex traffic scenarios, providing a basis for subsequent deployment-oriented studies on embedded or mobile platforms, pending dedicated efficiency benchmarking.

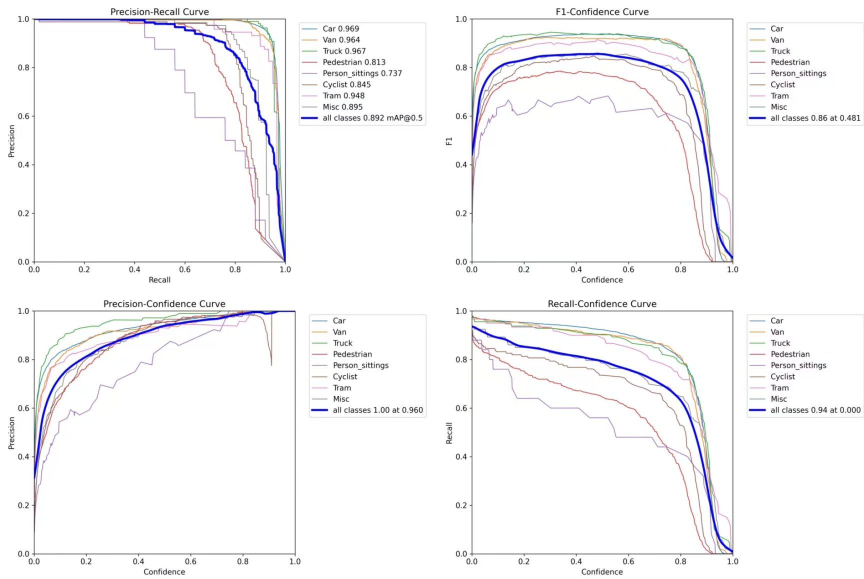

Fig. 4. Category-wise Performance Analysis: Precision-Recall Curves of the Improved Model

VI. Conclusion and Prospect

To address the challenge of balancing detection accuracy and real-time requirements in autonomous-driving scenarios, this study integrates a RepViT backbone into the YOLOv8 framework and proposes the RepViTStageYOLO module to align multi-scale features with the FPN+PAN pathway. Experimental results on the KITTI dataset show that the improved model achieves an mAP@0.5 of 0.892 and performs strongly on vehicle detection (Car AP = 0.969), with stable convergence behavior reflected by the loss curves. While the overall design is motivated by efficiency through structural re-parameterization, this paper focuses on accuracy evaluation and does not report efficiency metrics such as Params, FLOPs/GFLOPs, or FPS. Future work will provide comprehensive efficiency benchmarking on GPUs and embedded platforms and further improve small-object detection (e.g., Pedestrian/Cyclist) by incorporating attention mechanisms (SE/ECA), higher-resolution detection layers, and data augmentation strategies such as Copy-Paste.

References

[1] Jocher, G., Chaurasia, A. and Qiu, J., "YOLOv8: Ultralytics Next-Generation Object Detection," arXiv preprint arXiv:2301.07761, 2023.

[2] Ge, Z., Liu, S., Wang, F., Li, Z. and Sun, J., "YOLOX: Exceeding YOLO Series in 2021," arXiv preprint arXiv:2107.08430, 2021.

[3] Zhang, S., Chi, C., Yao, Y., Lei, Z. and Li, S.Z., "Bridging the Gap Between Anchor-Based and Anchor-Free Detection via Adaptive Training Sample Selection," 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, WA, USA, 2020, pp. 9759–9768.

[4] Tian, Z., Shen, C., Chen, H. and He, T., "FCOS: Fully Convolutional One-Stage Object Detection," 2019 IEEE/CVF International Conference on Computer Vision (ICCV), Seoul, Korea, 2019, pp. 9627–9636.

[5] Wang, C.H., Lin, Z., Han, J. and Ding, G., "RepViT: Revisiting Mobile CNN From ViT Perspective," 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, WA, USA, 2024, pp. 15909–15920.

[6] Lin, T.Y., Goyal, P., Girshick, R., He, K. and Dollár, P., "Focal Loss for Dense Object Detection," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 42, no. 2, pp. 318–327, Feb. 2020.

[7] He, K., Zhang, X., Ren, S. and Sun, J., "Deep Residual Learning for Image Recognition," 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Las Vegas, NV, USA, 2016, pp. 770–778.

[8] Khoramdel, J., Moori, A., Borhani, Y., Ghanbarzadeh, A. and Najafi, E., "YOLO-Former: YOLO Shakes Hand With ViT," arXiv preprint arXiv:2401.06244, 2024.

[9] Geiger, A., Lenz, P. and Urtasun, R., "Are We Ready for Autonomous Driving? The KITTI Vision Benchmark Suite," 2012 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Providence, RI, USA, 2012, pp. 3354–3361.

[10] Wang, Q., Wu, B., Zhu, P., Li, P., Zuo, W. and Hu, Q., "ECA-Net: Efficient Channel Attention for Deep Convolutional Neural Networks," 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, WA, USA, 2020, pp. 11531–11539.

[11] Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S. and Guo, B., "Swin Transformer: Hierarchical Vision Transformer Using Shifted Windows," 2021 IEEE/CVF International Conference on Computer Vision (ICCV), Montreal, QC, Canada, 2021, pp. 10012–10022.

[12] Ding, X., Zhang, X., Ma, N., Han, J., Ding, G. and Sun, J., "RepVGG: Making VGG-style ConvNets Great Again," 2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Nashville, TN, USA, 2021, pp. 13733–13742.

[13] Qi, Z., "Improved YOLOv5 Based on Attention Mechanism and FasterNet," arXiv preprint arXiv:2403.08499, 2024.

[14] Zhang, H., "MDCFVit-YOLO: Enhanced YOLOv8 for Nighttime Infrared Object Detection," Proceedings of the International Conference on Machine Vision, 2025.

[15] Zhong, J., "Improved Real-Time Object Detection Method Based on YOLOv8 with Faster-C2f and Rep Structures," Multimedia Tools and Applications, 2025.

[16] Xu, J., Pan, F., Han, X., Wang, L., Wang, Y. and Li, W., "EdgeTrim-YOLO: Improved Trim YOLO Framework Tailored for Deployment on Edge Devices," 2024 4th International Conference on Computer Communication and Artificial Intelligence (CCAI), Xi'an, China, 2024, pp. 113–118, doi:10.1109/CCAI61966.2024.10602964.

[17] Chen, Z., Yang, K., Wu, Y., Yang, H. and Tang, X., "HCLT-YOLO: A Hybrid CNN and Lightweight Transformer Architecture for Object Detection in Complex Traffic Scenes," IEEE Transactions on Vehicular Technology, vol. 74, no. 3, pp. 3681–3694, March 2025, doi:10.1109/TVT.2024.3496513.